



Physique Cours 8

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Génie Mécanique

1 Cinématique des référentiels accélérés

$$AP = \sum_i y_i \hat{y}_i$$

$$v_r(P) = \sum_i \dot{y}_i \hat{y}_i$$

$$v_a(P) = \frac{d}{dt}OP = \frac{d}{dt}OA + \frac{d}{dt}AP$$

$$v_a(P) = v_a(A) + v_r(P) + \Omega \wedge AP$$

$$a_a(P) = a_a(A) + a_{r(P)} + 2\Omega \wedge v_r(P) + \dot{\Omega} \wedge AP + \Omega \wedge (\Omega \wedge AP)$$

2 Angle d'Euler

On peut positionner un solide par 3 rotations successives

1. ψ autour de l'axe x_3
2. θ autour de l'axe x_1'
3. ϕ autour de l'axe x_3''

On a alors

1. $x_3 = x_3'$
2. $x_1' = x_1''$
3. $x_3'' = x_3'''$

$$\begin{pmatrix} x_1' \\ x_2' \\ x_3' \end{pmatrix} = \begin{pmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$\begin{pmatrix} x_1'' \\ x_2'' \\ x_3'' \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x_1' \\ x_2' \\ x_3' \end{pmatrix}$$

$$\begin{pmatrix} x_1''' \\ x_2''' \\ x_3''' \end{pmatrix} = \begin{pmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1'' \\ x_2'' \\ x_3'' \end{pmatrix}$$

$$\Omega = \dot{\psi} \hat{x}_3 + \dot{\theta} \hat{x}_1' + \dot{\phi} \hat{x}_3''$$

$$\Omega = \dot{\psi} \hat{x}_3' + \dot{\theta} \hat{x}_1'' + \dot{\phi} \hat{x}_3'''$$

$$\Omega = \begin{pmatrix} \dot{\psi} \sin(\theta) \sin(\phi) + \dot{\theta} \cos(\phi) \\ \dot{\psi} \sin(\theta) \cos(\phi) - \dot{\theta} \sin(\phi) \\ \dot{\psi} \cos(\theta) + \dot{\phi} \end{pmatrix}$$

3 Equations d'Euler

$$I_G = \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix}$$

$$I_1 \dot{\Omega}_1 = (I_2 - I_3) \Omega_2 \Omega_3 + M_{G1}$$

$$I_2 \dot{\Omega}_2 = (I_3 - I_1) \Omega_1 \Omega_3 + M_{G2}$$

$$I_3 \dot{\Omega}_3 = (I_1 - I_2) \Omega_1 \Omega_2 + M_{G3}$$

$$\frac{d(L_G)}{dt} = M_G$$

$$L_G = I_G \Omega$$

4 Conservation de l'énergie cinétique

$$E_c = \frac{1}{2} \Omega^T I_G \Omega$$

$$E_c = \frac{1}{2} (I_1 \Omega_1^2 + I_2 \Omega_2^2 + I_3 \Omega_3^2)$$

$$\frac{d}{dt} E_c = I_1 \Omega_1 \dot{\Omega}_1 + I_2 \Omega_2 \dot{\Omega}_2 + I_3 \Omega_3 \dot{\Omega}_3$$

$$\frac{d}{dt} E_c = ((I_2 - I_3) + I_3 - I_1 + I_1 - I_2) \Omega_1 \Omega_2 \Omega_3$$

$$\frac{d}{dt} E_c = 0$$

5 Conservation du module de L_G

$$L_G = \sum_i I_i \Omega_i \hat{y}_i = I_G \Omega$$

$$\|L_G\|^2 = I_G \Omega \cdot I_G \Omega = I_1^2 \Omega_1^2 + I_2^2 \Omega_2^2 + I_3^2 \Omega_3^2$$

$$\frac{d}{dt} \|L_G\|^2 = 0$$

Intersection de deux ellipsoïdes